Nonexistence of consecutive powerful triplets around cubes with mixed prime factorizations

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September 2026

Abstract

A positive integer is *powerful* if every prime in its factorization occurs with exponent at least two. The Erdős–Mollin–Walsh conjecture asserts that no three consecutive integers are all powerful. Chan (2025) excluded triples of the form $x^3-1=p^3y^2,\ x^3,\ x^3+1=q^3z^2$, and She (2025) excluded $x^3-1=p^2a^3,\ x^3,\ x^3+1=q^2b^3$, with p,q prime. We close the remaining *mixed* configurations: there are no consecutive powerful triples of the form $x^3-1=p^2a^3,\ x^3+1=q^3z^2$, nor of the form $x^3-1=p^3y^2,\ x^3+1=q^2b^3$. The case analysis reduces both families to the single Diophantine equation $t^6+t^3+1=3w^2$, which we solve completely: exploiting the palindromic symmetry of the genus-2 curve $3w^2=t^6+t^3+1$, we descend to an elliptic quotient of Mordell–Weil rank 0 (curve `1296.k1` in the LMFDB, Cremona label `1296f2`), so the only rational solutions are $(t,w)=(1,\pm1)$. We further show that no finite congruence sieve can decide this equation, so that some global (non-congruence) argument is required. As a complementary result we refine Beckon's mod-36 constraint on the smallest member of a hypothetical triple to arbitrary prime-square moduli and quantify the density decay of admissible residue classes.

1 Introduction

A positive integer m is *powerful* if $p^2 \mid m$ for every prime $p \mid m$; equivalently $m=a^2b^3$ with $a,b\in\mathbb{Z}_{>0}$ [7]. Pairs of consecutive powerful numbers, such as $(8,9)$ and $(288,289)$, occur infinitely often (e.g. via the Pell equation $x^2-8y^2=1$). In contrast, Erdős [5] conjectured, and Mollin and Walsh [9, 10] investigated systematically, that *no three consecutive integers are all powerful.* The conjecture remains open; under the *abc*-conjecture at most finitely many triples exist.

If $(n,n+1,n+2)$ is such a triple, the middle member may or may not be a perfect cube. The case in which the middle member is a cube x^3 has attracted recent attention. Chan [4] proved that there is no triple of the shape

$$x^3-1=p^3y^2,\qquad x^3,\qquad x^3+1=q^3z^2\qquad (p,q \text{ prime}),$$

and She [12] proved the analogous statement for

$$x^3-1=p^2a^3,\qquad x^3,\qquad x^3+1=q^2b^3\qquad (p,q \text{ prime}).$$

Both works treat *symmetric* configurations: the two outer members carry the same shape ($p^3 \cdot \square$ on both sides in [4]; $p^2 \cdot$ cube on both sides in [12]). Further recent activity in this circle includes van Doorn's study of three-term arithmetic progressions of consecutive powerful numbers [15]. The natural *mixed* configurations have, to our knowledge, not been treated. We close them.

Theorem 1. *There are no integers $x \geq 2$, primes p, q, and integers a, b, y, z such that either*

$$\text{(M1)} \qquad x^3 - 1 = p^2 a^3 \quad \textit{and} \quad x^3 + 1 = q^3 z^2, \tag{1}$$

$$\text{(M2)} \qquad x^3 - 1 = p^3 y^2 \quad \textit{and} \quad x^3 + 1 = q^2 b^3. \tag{2}$$

In particular, no triple of consecutive powerful numbers $(x^3 - 1,\, x^3,\, x^3 + 1)$ has mixed outer factorizations of these shapes.

The proof is a case analysis in the spirit of [4, 12] (Sections 3–4): factoring $x^3 \mp 1 = (x \mp 1)(x^2 \pm x + 1)$ and tracking 3-adic valuations reduces all but one branch of each family to classical Mordell equations $Y^2 = X^3 + k$ with $k \in \{2, -2, 54, -54, -162\}$, whose integral points are completely known [6, 2, 8]. Both remaining branches lead to one and the same equation,

$$t^6 + t^3 + 1 = 3w^2, \tag{3}$$

with $t \geq 7$ in family (M1) and $t \leq -5$ in family (M2). Equation (3) concerns the ninth cyclotomic value $\Phi_9(t)$ and defines a curve of genus 2, outside the reach of the degree-4 reductions used in [4, 12]; moreover, we prove (Remark 14) that *no* finite congruence sieve can decide it. Instead we solve it completely:

Theorem 2. *The only solutions of $t^6 + t^3 + 1 = 3w^2$ in rational numbers are $(t, w) = (1, \pm 1)$.*

The proof (Section 5) exploits the palindromic symmetry $t \mapsto 1/t$ of the genus-2 curve $3w^2 = t^6 + t^3 + 1$: the curve admits two independent elliptic quotients over $\mathbb{Q}$, and one of them has Mordell–Weil rank 0 with trivial torsion — it is the curve `1296.k1` in the LMFDB [8], Cremona label `1296f2` — which pins down all rational points. Since $t = 1$ lies in neither branch range, both remaining branches are empty and Theorem 1 follows. Figure 1 summarizes the four shape combinations and the reduction.

As a complementary result in the same circle of ideas we record a modular refinement of a constraint of Beckon [1] on arbitrary consecutive powerful triples (not assuming a cube in the middle).

Proposition 3. *Let $(n, n+1, n+2)$ be powerful. Then, beyond Beckon's constraint $n \bmod 36 \in \{7, 27, 35\}$:*

(i) *$n \bmod 900$ lies in an explicit set of 39 residue classes, and $n \bmod 44100$ in an explicit set of 1209 classes;*

(ii) *for every prime $\ell \geq 5$, the proportion of admissible residues modulo ℓ^2 is $(\ell^2 - 3\ell + 3)/\ell^2$, while it is $1/4$ for $\ell = 2$ and $1/3$ for $\ell = 3$;*

(iii) *consequently the density of admissible classes modulo $\prod_{\ell \leq L} \ell^2$ tends to 0 as $L \to \infty$, but only at the rate $\asymp (\log L)^{-3}$; in particular no finite sieve of this type can resolve the Erdős–Mollin–Walsh conjecture.*

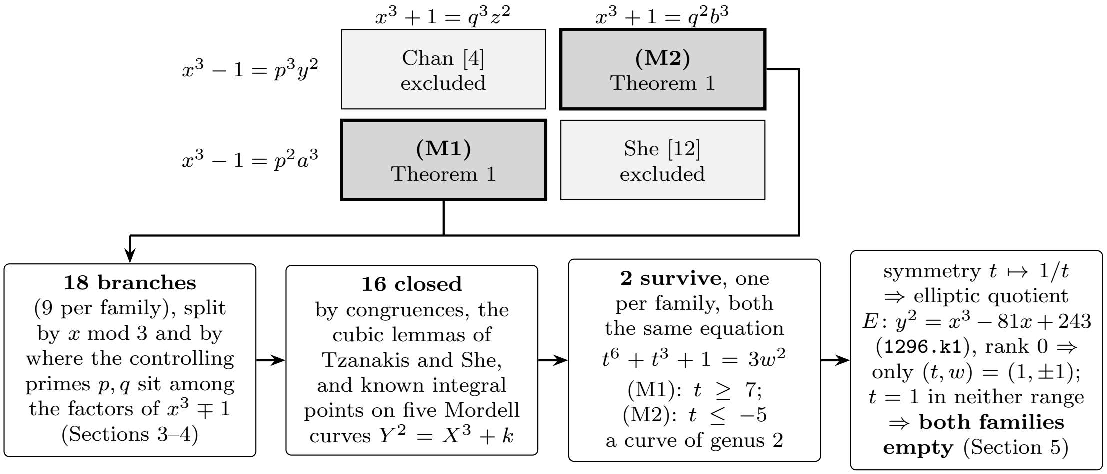

Figure 1: Overview. Top: the four shape combinations for the outer members of a consecutive powerful triple ($x^3 - 1$, x^3, $x^3 + 1$) with p, q prime; the symmetric cases were excluded by Chan [4] and She [12], the mixed cases (M1), (M2) by Theorem 1. Bottom: how the mixed cases close. A branch is counted once for each case marked *Closed* in Sections 3–4; the five Mordell curves are those of Lemma 11.

All computer calculations below are short and reproducible; the elliptic-curve computations were performed in SageMath and independently cross-checked (two independent runs, plus the LMFDB data [8]); the symbolic identities were verified exactly in SymPy. The verification scripts are provided as ancillary files.

2 Preliminaries

Throughout, $x \geq 2$ is an integer, p, q are primes, and v_ℓ denotes the ℓ-adic valuation. We collect standard facts; proofs are short and included for completeness where not cited.

Lemma 4 (parity)**.** *In any triple of consecutive powerful numbers the even member is divisible by* 4*. If the middle member is x^3, then x is even; in particular $x^3 \pm 1$ are odd and coprime, so $p \neq q$ in* (1)*–*(2)*.*

Proof. Exactly one or two of three consecutive integers are even; a powerful even number is $\equiv 0 \pmod 4$. If n and $n + 2$ were both even, one of them would be $\equiv 2 \pmod 4$, hence not powerful. Thus the even member is unique, namely the middle one if the outer two are odd. For the middle member x^3: if x were odd, then $x^3 \pm 1$ would be even, contradiction. Finally $\gcd(x^3 - 1, x^3 + 1) \mid 2$ and both are odd. □

Lemma 5 (gcd and LTE)**.** $x^3 \mp 1 = (x \mp 1)(x^2 \pm x + 1)$ *with* $g_\mp := \gcd(x \mp 1, x^2 \pm x + 1) = \gcd(x \mp 1, 3) \in \{1, 3\}$*. Moreover, if* $3 \mid x \mp 1$ *then* $v_3(x^2 \pm x + 1) = 1$ *exactly.*

Proof. $x^2 \pm x + 1 = (x \mp 1)(x \pm 2) + 3$, giving the gcd claim (cf. [12, Lemma 4]). For the valuation: if $x = 1 + 3k$ then $x^2 + x + 1 = 3 + 9k + 9k^2 \equiv 3 \pmod 9$; if $x = -1 + 3k$ then $x^2 - x + 1 = 3 - 9k + 9k^2 \equiv 3 \pmod 9$. In both cases $v_3 = 1$ exactly. □

Lemma 6 (square sandwich). *For $x \geq 2$: $x^2 < x^2 + x + 1 < (x+1)^2$ and $(x-1)^2 < x^2 - x + 1 < x^2$. Hence neither $x^2 + x + 1$ nor $x^2 - x + 1$ is a perfect square.*

Lemma 7 (Chan-side structure). *Suppose $x^3 + 1 = q^3 z^2$ and $g_+ = 1$. Then $x+1$ is a perfect square and $x^2 - x + 1 = q^{3+2k} \cdot \square$ for some $k \geq 0$. Analogously, if $x^3 - 1 = p^3 y^2$ and $g_- = 1$, then $x - 1$ is a perfect square and $x^2 + x + 1 = p^{3+2k} \cdot \square$.*

Proof. With $g_+ = 1$ the factors $x+1$ and $x^2 - x + 1$ are coprime, so each is, up to the prime q, a perfect square, and q^3 (odd exponent ≥ 3) divides exactly one of them. If q^3 divided $x+1$ with $x^2 - x + 1$ a square, Lemma 6 is contradicted. The mirrored statement is identical. $\square$

Lemma 8 (She's splitting; [12, Lemma 1]). *If $RS = p^2 C^3$ with $\gcd(R, S) = 1$, then one of R, S is a perfect cube and the other is p^2 times a perfect cube.*

Lemma 9 ([12, Lemma 2]). *$u^2 + u + 1 = 3v^3$ has only the integer solutions $u \in \{-2, 1\}$, and $u^2 - u + 1 = 3v^3$ only $u \in \{-1, 2\}$.*

Lemma 10 (Tzanakis; [14], cf. [12, Lemma 3]). *$u^2 + u + 1 = v^3$ has only $u \in \{-19, -1, 0, 18\}$, and $u^2 - u + 1 = v^3$ only $u \in \{-18, 0, 1, 19\}$.*

Lemma 11 (Mordell curves). *The integral points of $Y^2 = X^3 + k$ are: for $k = 2$: $(-1, \pm 1)$; for $k = -2$: $(3, \pm 5)$; for $k = 54$: $(3, \pm 9)$; for $k = -54$: $(7, \pm 17)$; for $k = -162$: none.*

Proof. All five values satisfy $|k| \leq 10^4$ and are covered by the complete solution of Mordell's equation in this range [6], extended to $|k| \leq 10^7$ in [2]; see also the corresponding LMFDB data [8]. (The case $k = -2$ is classical.) We re-verified all five lists with `integral_points` in SageMath in two independent runs. $\square$

We will also use without further comment: squares mod 9 are $\{0, 1, 4, 7\}$; cubes mod 9 are $\{0, 1, 8\}$; cubes of integers $\equiv 1 \pmod 3$ are $\equiv \{1, 10, 19\} \pmod{27}$.

Exponent bookkeeping. For the Chan shape $x^3 \mp 1 = P^3 Y^2$ one has $v_P \equiv 1 \pmod 2$, $v_P \geq 3$, and $v_r \equiv 0 \pmod 2$ for primes $r \neq P$; for the She shape $x^3 \mp 1 = P^2 A^3$ one has $v_P \equiv 2 \pmod 3$ and $v_r \equiv 0 \pmod 3$ for $r \neq P$. Since $g_\mp \in \{1, 3\}$, every prime $r \neq 3$ divides exactly one of the two factors $x \mp 1$, $x^2 \pm x + 1$.

3 Family (M1): $x^3 - 1 = p^2 a^3$, $x^3 + 1 = q^3 z^2$

By Lemma 4, x is even and $p \neq q$. We split by x mod 3.

Case 1: $x \equiv 0 \pmod 3$ (so $g_- = g_+ = 1$)

By Lemma 7, $x + 1 = v^2$. By Lemma 8 applied to $(x-1)(x^2 + x + 1) = p^2 a^3$:

- (ii) $x - 1 = p^2 u^3$, $x^2 + x + 1 = t^3$: Lemma 10 forces $x \in \{-19, -1, 0, 18\}$; only $x = 18$ is even and ≥ 2, but $x - 1 = 17$ is prime, not of the form $p^2 u^3$. *Closed.*
- (i) $x - 1 = u^3$, $x^2 + x + 1 = p^2 t^3$: combining with $x + 1 = v^2$ gives $v^2 - u^3 = 2$, a point on $Y^2 = X^3 + 2$; by Lemma 11 $u = -1$, $x = 0 < 2$. *Closed.*

Case 2: $x \equiv 1 \pmod 3$ **(so** $g_- = 3$**,** $g_+ = 1$**)**

Here $x^2 - x + 1 \equiv 1 \pmod 3$, so $q \neq 3$, and $x + 1 = v^2$ by Lemma 7; on the She side $v_3(x^2 + x + 1) = 1$ exactly (Lemma 5).

- $p = 3$: here $v_3(x^3 - 1) = 2 + 3v_3(a)$, while $v_3(x^3 - 1) = v_3(x - 1) + 1$ by Lemma 5; hence $v_3(x - 1) = 1 + 3v_3(a)$ and $v_3(x^2 + x + 1) = 1$. All prime exponents in $x^2 + x + 1$ other than that of 3 are $\equiv 0 \pmod 3$, so $x^2 + x + 1 = 3t^3$ with $3 \nmid t$; Lemma 9 gives $x \in \{-2, 1\}$, both < 2. *Closed.*

- $p \neq 3$, $p \mid x - 1$: then again $x^2 + x + 1 = 3t^3$ ($v_3 = 1$, $p \nmid$, all other exponents $\equiv 0 \bmod 3$), so $x \in \{-2, 1\}$. *Closed.*

- $p \neq 3$, $p \mid x^2 + x + 1$: then $v_3(x - 1) = 3v_3(a) - 1 \equiv 2 \pmod 3$ and all other prime exponents in $x - 1$ are $\equiv 0 \pmod 3$, hence $x - 1 = 9d^3$. But $x + 1 = v^2$ gives $v^2 = 9d^3 + 2 \equiv 2 \pmod 9$, impossible since squares mod 9 are $\{0, 1, 4, 7\}$. *Closed.*

Case 3: $x \equiv 2 \pmod 3$ **(so** $g_- = 1$**,** $g_+ = 3$**)**

She side (Lemma 8):

(ii) $x - 1 = p^2u^3$, $x^2 + x + 1 = t^3$: Lemma 10 gives $x \in \{-19, -1, 0, 18\}$; none is $\equiv 2 \pmod 3$, even, and ≥ 2. *Closed.*

(i) $x - 1 = u^3$ with u odd, $u \equiv 1 \pmod 3$ (so $u \equiv 1 \bmod 6$), and $x^2 + x + 1 = p^2t^3$. On the Chan side $g_+ = 3$ and $v_3(x^2 - x + 1) = 1$ exactly. Sub-split on q:

- $q = 3$: $x^3 + 1 = 27z^2$ gives $v_3(x + 1) = 2 + 2v_3(z)$ even and all other exponents in $x + 1$ even, so $x + 1 = m^2$; then $m^2 - u^3 = 2$, again $Y^2 = X^3 + 2$, so $u = -1$, $x = 0 < 2$. *Closed.*
- $q \neq 3$, $q^3 \mid x^2 - x + 1$: then $v_3(x + 1)$ is odd, say $x + 1 = 3^{2j+1}m^2$ with $\gcd(m, 3) = 1$ up to squares. If $j \geq 1$ then $27 \mid x + 1 = u^3 + 2$, i.e. $u^3 \equiv -2 \equiv 25 \pmod{27}$, impossible for $u \equiv 1 \pmod 3$. Hence $j = 0$, $x + 1 = 3m^2$, i.e. $u^3 = 3m^2 - 2$; multiplying by 27: $(9m)^2 = (3u)^3 + 54$. By Lemma 11 the only points have $X = 3u = 3$, so $u = 1$, $x = 2$; but $x^3 - 1 = 7$ is not powerful. *Closed.*
- $q \neq 3$, $q^3 \mid x + 1$: then $x^2 - x + 1 = 3w^2$ ($v_3 = 1$ exactly, $q \nmid$, all other exponents even). With $x = u^3 + 1$:

$$(u^3 + 1)^2 - (u^3 + 1) + 1 = u^6 + u^3 + 1 = 3w^2,$$

i.e. equation (3) with $t = u$. Here $u \geq 1$ since $x \geq 2$, and $u = 1$ gives $x = 2$, where $x^3 - 1 = 7$ is not of the form p^2a^3; together with $u \equiv 1 \pmod 6$ this forces $t = u \geq 7$. *This is the only surviving branch of (M1); it is empty by Theorem 2.*

4 Family (M2): $x^3 - 1 = p^3y^2$, $x^3 + 1 = q^2b^3$

Again x is even, $p \neq q$; split by x mod 3.

Case 1: $x \equiv 0 \pmod 3$ **(**$g_- = g_+ = 1$**)**

Chan side (Lemma 7): $x - 1 = u^2$ with u odd. She side (Lemma 8) applied to $(x+1)(x^2 - x + 1) = q^2 b^3$:

(b) $x + 1 = q^2 s^3$, $x^2 - x + 1 = t^3$: Lemma 10 gives $x \in \{-18, 0, 1, 19\}$; no admissible value. *Closed.*

(a) $x + 1 = s^3$, $x^2 - x + 1 = q^2 t^3$: then $s^3 - u^2 = 2$, i.e. a point on $Y^2 = X^3 - 2$; by Lemma 11 $(s, u) = (3, \pm 5)$, $x = 26 \equiv 2 \pmod 3$, contradicting the case. *Closed.*

Case 2: $x \equiv 1 \pmod 3$ **(**$g_- = 3$**,** $g_+ = 1$**)**

She side has $g_+ = 1$:

(b) $x + 1 = q^2 s^3$, $x^2 - x + 1 = t^3$: as above, no admissible x. *Closed.*

(a) $x+1 = s^3$ with s odd, $s \equiv 2 \pmod 3$, hence $s \equiv 5 \pmod 6$ and $s \geq 5$; and $x^2 - x + 1 = q^2 t^3$. On the Chan side $v_3(x^2 + x + 1) = 1$ exactly. Sub-split on p:

- $p = 3$: then $v_3(x-1)$ is even and all other exponents in $x-1$ are even, so $x - 1 = u^2$; then $u^2 = s^3 - 2$, so $(s, u) = (3, \pm 5)$ by Lemma 11, giving $x = 26 \equiv 2 \pmod 3 \neq 1$. *Closed.*
- $p \neq 3$, $p \mid x - 1$: then $x^2 + x + 1 = 3w^2$ ($v_3 = 1$, $p \nmid$, rest even). With $x = s^3 - 1$:

$$(s^3 - 1)^2 + (s^3 - 1) + 1 = s^6 - s^3 + 1 = 3w^2,$$

i.e. equation (3) with $t = -s \leq -5$. *This is the only surviving branch of (M2); it is empty by Theorem 2.*
- $p \neq 3$, $p \mid x^2 + x + 1$: then $v_3(x - 1)$ is odd, $x - 1 = 3^{2j+1} m^2$. If $j \geq 1$ then $27 \mid s^3 - 2$, so $s^3 \equiv 2 \pmod 9$, impossible (cubes mod 9). Hence $j = 0$, $s^3 = 3m^2 + 2$; multiplying by 27: $(9m)^2 = (3s)^3 - 54$ with $3 \mid X = 3s$. By Lemma 11 the only integral points have $X = 7$, not divisible by 3. *Closed.*

Case 3: $x \equiv 2 \pmod 3$ **(**$g_- = 1$**,** $g_+ = 3$**)**

Chan side: $x - 1 = u^2$ (u odd), $x^2 + x + 1 = p^{3+2k} \cdot \square$. She side: $v_3(x^2 - x + 1) = 1$ exactly. We first dispose of the case $q = 3$:

- $q = 3$: here $x^3 + 1 = 9b^3$, so $v_3(x^3 + 1) = 2 + 3v_3(b)$, while $v_3(x^3 + 1) = v_3(x + 1) + 1$ by Lemma 5; hence $v_3(x + 1) = 1 + 3v_3(b)$ and $v_3(x^2 - x + 1) = 1$, with all other prime exponents in $x^2 - x + 1$ being $\equiv 0 \pmod 3$. Thus $x^2 - x + 1 = 3s^3$, and Lemma 9 gives $x \in \{-1, 2\}$; $x = 2$ fails since $x^3 - 1 = 7$ is not of the form $p^3 y^2$. *Closed.*

- $q \neq 3$: from $x^3 + 1 = q^2 b^3$ and $v_3(x^2 - x + 1) = 1$ we get $v_3(x + 1) = 3v_3(b) - 1 \equiv 2 \pmod 3$, in particular $9 \mid x + 1$. Sub-split:

  - $q \mid x + 1$: then $x^2 - x + 1 = 3t^3$ ($v_3 = 1$, $q \nmid$, all other exponents $\equiv 0 \bmod 3$), so $x \in \{-1, 2\}$ by Lemma 9; as before. *Closed.*

- $q \mid x^2 - x + 1$: then $x + 1 = 3^{3v_3(b)-1} \cdot s^3 = 9c^3$ for an integer c, and $x - 1 = u^2$ gives $u^2 + 2 = 9c^3$; multiplying by 81: $(9u)^2 = (9c)^3 - 162$. By Lemma 11, $Y^2 = X^3 - 162$ has no integral points at all. *Closed.*

This completes the case analysis: both families reduce to the branches recorded above, all of which are empty except the two leading to (3).

5 The equation $t^6 + t^3 + 1 = 3w^2$

Write $\Phi_9(t) = t^6 + t^3 + 1$ for the ninth cyclotomic polynomial, so (3) reads $\Phi_9(t) = 3w^2$. The smooth projective model of

$$\mathcal{C} : \; Y^2 = 3t^6 + 3t^3 + 3 \qquad (Y = 3w)$$

is a curve of genus 2. Since the right-hand side is palindromic, $\mathcal{C}$ carries the involution $\sigma : (t, Y) \mapsto (1/t, \, Y/t^3)$ defined over $\mathbb{Q}$, and hence (besides the hyperelliptic involution ι) the second involution $\tau = \sigma\iota$. The quotients $\mathcal{C}/\sigma$ and $\mathcal{C}/\tau$ are elliptic, and $\mathrm{Jac}(\mathcal{C})$ is isogenous over $\mathbb{Q}$ to their product. Concretely:

Lemma 12. *Set $u = t + 1/t$, and*

$$V = \frac{Y(t+1)}{t^2}, \qquad W = \frac{Y(t-1)}{t^2}.$$

Then on $\mathcal{C}$ one has the exact identities

$$V^2 = 3(u+2)(u^3 - 3u + 1) =: q_+(u), \qquad W^2 = 3(u-2)(u^3 - 3u + 1) =: q_-(u).$$

Proof. Multiply $Y^2 = 3(t^6 + t^3 + 1)$ by $(t \pm 1)^2/t^4$ and use

$$(t \pm 1)^2 \, (t^6 + t^3 + 1) = t^4 \, (u \pm 2) \, (u^3 - 3u + 1),$$

which is a polynomial identity in t after substituting $u = t + 1/t$ (verified exactly; it follows from $u^3 - 3u + 2 = (u-1)^2(u+2)$ and $u^3 - 3u - 2 = (u+1)^2(u-2)$ together with $t^3 + t^{-3} = u^3 - 3u$ and $(t \pm 1)^2/t = u \pm 2$). $\square$

Proof of Theorem 2. Let $(t, w) \in \mathbb{Q}^2$ satisfy (3) and put $Y = 3w$. Note $t \neq 0$ (else $3w^2 = 1$) and $u = t + 1/t \neq -1$ (else $t^2 + t + 1 = 0$, no rational root). By Lemma 12, (u, W) is a rational point on the genus-one quartic curve

$$\mathcal{D}_- : \; W^2 = q_-(u) = 3(u-2)(u^3 - 3u + 1).$$

The quartic q_- is squarefree (its cubic factor has discriminant 81 and does not vanish at $u = 2$), so $\mathcal{D}_-$ is smooth of genus one; its two points at infinity satisfy $W_\infty^2 = 3$ and are conjugate over $\mathbb{Q}(\sqrt{3})$, hence not rational. $\mathcal{D}_-$ has the rational point $(2, 0)$.

We claim $\mathcal{D}_-(\mathbb{Q}) = \{(2, 0)\}$. Substituting $u = 2 + 1/T$, $W = S/T^2$ transforms $\mathcal{D}_-$ birationally into

$$S^2 = 9T^3 + 27T^2 + 18T + 3,$$

and the further substitution $(x,y) = (9T+9,\, 9S)$ gives the elliptic curve

$$E:\ y^2 = x^3 - 81x + 243,$$

of conductor 1296. This is the curve `1296.k1` in the LMFDB [8] (Cremona label `1296f2`): it has Mordell–Weil rank 0 and trivial torsion, so $E(\mathbb{Q}) = \{\mathcal{O}\}$ — there is no affine rational point. (We verified rank 0 and triviality of torsion independently in SageMath with `rank(proof=True)`.) Explicitly, the maps are inverted by $T = (x-9)/9$, $S = y/9$, so that $u = 2 + 1/T = (2x-9)/(x-9)$ and $W = S/T^2 = 9y/(x-9)^2$; under this correspondence the affine rational points of $\mathcal{D}_-$ with $u \neq 2$ are in bijection with the affine rational points of E with $x \neq 9$. No rational points are lost at the exceptional loci: the two points at infinity of $\mathcal{D}_-$ are irrational (as noted above, $W_\infty^2 = 3$), the point $(2,0)$ corresponds to $\mathcal{O} \in E$, and the points of E with $x = 9$ satisfy $y^2 = 243$ and are likewise irrational. Since $E(\mathbb{Q}) = \{\mathcal{O}\}$ contains no affine point, $u = 2$ is the only possibility, and then $W^2 = q_-(2) = 0$.

Pulling back to $\mathcal{C}$: $u = t + 1/t = 2$ forces $(t-1)^2 = 0$, so $t = 1$ and $Y^2 = 9$, $w = \pm 1$. □

Remark 13. The companion quotient $\mathcal{D}_+ : V^2 = q_+(u)$ has Jacobian $y^2 = x^3 - 189x + 999$ of rank 1 (LMFDB `1296.b1`, Cremona `1296l2`); this is why the descent must go through $\mathcal{D}_-$.

Remark 14 (no congruence sieve suffices). Equation (3) cannot be resolved by congruences alone, in the following precise sense. Substituting $z = t^3$, any solution gives a point on the conic $z^2 + z + 1 = 3w^2$; writing $2z + 1 = 3c$, $2w = b$ reduces the conic to the Pell equation $b^2 - 3c^2 = 1$, whose solutions with b even form a single orbit $\{z_k\}_{k\in\mathbb{Z}}$ of the recurrence $z_{k+1} = 14z_k - z_{k-1} + 6$ through $(z_0, z_1) = (-2, 1)$, the sign choice corresponding to the symmetry $z \mapsto -1 - z$. Asking whether (3) has a solution with $t \neq 1$ is asking whether the orbit contains a cube other than $z_1 = 1$. Since $z_1 = 1$ *is* a cube, its residue class modulo any period of the sequence is a "shadow" that no congruence can eliminate — and via the symmetry it also protects a class in the negative branch. Hence no finite system of congruence conditions can prove the nonexistence of further cubes in the orbit, and some global (here: elliptic) input is unavoidable. A direct check of the recurrence (in exact integer arithmetic) confirms that no z_k with $k \leq 2000$ other than z_1 is a cube, the largest candidates having more than 2200 decimal digits. (This computation only illustrates the covering obstruction; it is not used in the proof of Theorem 2, which rests entirely on the rank-0 descent of this section.)

Remark 15 (relation to Nagell–Ljunggren and to powers in recurrences). Since $\Phi_9(t) = (t^9 - 1)/(t^3 - 1)$, equation (3) is adjacent to the classical equations $(x^n - 1)/(x - 1) = 3y^m$ of Nagell–Ljunggren type. Note however that Nagell's theorem on $x^2 + x + 1 = 3y^n$ $(n > 2)$ does not apply: in (3) the square sits on the right-hand side ($m = 2$), while the cube condition sits inside the conic variable $z = t^3$. In the language of Remark 14, (3) asks for the cubes in a fixed binary recurrence orbit; the effective methods underlying the finiteness results of Pethő [11] and Shorey–Stewart [13] guarantee that there are only finitely many such cubes, but do not by themselves produce the explicit solution set. For the closely related factorization $(2z+1)(z^2+z+1) = y^\ell$ see Bennett, Patel, and Siksek [3]. We are not aware of a published complete determination of the integer (or rational) solutions of (3); the contribution here is the explicit resolution via the rank-0 elliptic quotient.

6 A modular refinement of Beckon's condition

Beckon [1] observed that the smallest member n of a triple of consecutive powerful numbers must satisfy $n \equiv 7, 27, 35 \pmod{36}$; this elementary but useful constraint appears not to have been taken up in the recent literature. The underlying mechanism generalizes to any prime-square modulus.

Proof of Proposition 3. If m is powerful and ℓ prime, then $v_\ell(m) \neq 1$; in particular, if $m \equiv c \pmod{\ell^2}$ with $\ell \mid c$, $\ell^2 \nmid c$, then m is not powerful. Call $r \bmod \ell^2$ *admissible* if none of $r, r+1, r+2$ is $\equiv \ell c' \pmod{\ell^2}$ with $\ell \nmid c'$. For $\ell \geq 5$ the $\ell-1$ forbidden residues $\ell c'$ are pairwise at distance ≥ 5, so the $3(\ell-1)$ shifted prohibitions are distinct, leaving $\ell^2 - 3(\ell-1) = \ell^2 - 3\ell + 3$ admissible classes. For $\ell = 2$: n must be odd and $n+1 \equiv 0 \pmod 4$, leaving the class $3 \bmod 4$ (1 of 4). For $\ell = 3$: the windows avoiding $\{3, 6\} \bmod 9$ start at $r \in \{0, 7, 8\}$ (3 of 9). By CRT the admissible classes multiply: modulo 36 one obtains exactly Beckon's $\{7, 27, 35\}$; modulo $900 = 2^2 3^2 5^2$ there are $3 \cdot 13 = 39$ classes; modulo $44100 = 2^2 3^2 5^2 7^2$ there are $39 \cdot 31 = 1209$. (The explicit lists were computed by two independent implementations with matching output.) Finally, $\prod_{5 \leq \ell \leq L} \left(1 - \frac{3}{\ell} + \frac{3}{\ell^2}\right) \asymp (\log L)^{-3}$ by Mertens' theorem, which proves (iii). $\square$

7 Concluding remarks

Together with [4] and [12], Theorem 1 completes the four shape-combinations $\{p^3 \cdot \square, p^2 \cdot \text{cube}\}^2$ for consecutive powerful triples around a cube in which both outer members are divisible by a single controlling prime to an exact pattern. Natural next steps in this program include the relaxations $p = q$ and composite cofactors, and She's question whether $x^n - 1$ can be powerful for $x > 1$, $n > 2$. The bielliptic reduction of Section 5 may be of independent use for other palindromic sextic obstructions in this circle of problems.

Use of AI tools

In the research, processing, and writing of this paper and its results I worked together with generative AI tools (large language models, mainly Claude, by Anthropic, inside Claude Code; the mathematical work in this note was done with the Fable 5 model). For this paper the mathematical work itself ran essentially through a single instance. I built and refined a supporting harness for mathematics that the instance can access, a set of rules and parameters that I improved and tested with agents over many rounds, benchmarking models and configurations against the sub-cases that were still open, in order to find out which setup actually made progress where earlier ones had stalled, and feeding what I learned back into the rules. Verification was then run separately, through independent checking passes whose task was to attack the result rather than to produce it. In other work I moved to a system of several coordinated specialized instances, but that is not what produced this note: one instance was enough. I lead this collaboration: I choose the research directions, set the goals, and make the final decisions in open exchange with the AI, learning actively as the work proceeds. The AI carries out the drafting, including the mathematical and technical parts, the numerical computation, and the literature search, under my direction. The AI works autonomously only task by task, within the structure I develop through feedback: it completes a task, and at open questions that need me it stops until the point is settled before the next step. Along the way I witness and take the decisions that shape the path, and it is usual for me to spot things that need improvement or closer thought. The work spans many separate runs over several weeks; and had I let the AI do all of it together alone, even if it is possible, it would no longer be my work but the AI's.

For this paper in particular I want to be precise about who did what, since the mathematics is the substance of the result. The case analysis of Sections 3–4 and the reduction of Section 5, including the descent to the rank-zero elliptic quotient, were produced by the AI under my direction. My own contribution was to choose the target, to work towards it by driving the improvements and the research it required, to keep probing the remaining gaps until they closed, to reject the attempts that did not hold, and to require and organize the verification. I do not claim the mathematical invention here as my own work.

I run multiple verifications at the different stages of the work and one before release, including cross-checks with an unrelated AI model from a different company, and all references are checked against the original sources. Concretely, the symbolic identities are confirmed exactly in SymPy, and the Mordell–Weil rank in SageMath (`integral_points`, `rank` with `proof=True`, in two independent runs) and against the LMFDB. In the end what matters are human eyes, a principle that is itself written into the parameters of my system: I have to reach out to experts after publishing for review and feedback, so I learn what is solid and what must be corrected or falsified. My scripts for reproduction and review are released with this record. These tools are not authors; I am the author, and I take full responsibility for all scientific content and decisions leading to these results and their publication.

Acknowledgements

No external funding was received. The author declares no competing interests.

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